

ASSIGNMENT

INDIVIDUAL CONTRIBUTION REPORT

My individual contribution to the **EV ChargeHub – Smart Campus Charging Management System** focused on the development and organization of the core application functionality, data management, charging workflow, validation, payment processing, dashboard integration, and testing. The main objective of my work was to ensure that the different modules of the application were logically connected and could support the complete charging-management process.

I contributed to the design and organization of the major entities used in the system, including User, Vehicle, Station, Point, Reservation, Session, and Payment. These entities represent the important components of a campus-based EV charging environment. I worked on understanding the relationship between these objects and how information moves between different modules. For example, users are associated with vehicles, stations contain charging points, reservations are associated with vehicles and charging points, and completed charging sessions are connected with payment records. This object-oriented organization helped maintain a structured design throughout the application.

I also contributed to the charging-station and charging-point management functionality. The system provides an interface for displaying charging stations, individual charging points, connector information, and operational status. I worked on the workflow through which charging-point status can be viewed and updated. The updated status is reflected in the relevant parts of the application so that the operator can obtain a clear view of the current charging infrastructure. This functionality is important because the availability of a charging point directly affects reservation and charging operations.

Another major area of my contribution was the reservation workflow. I worked on the process of selecting a registered vehicle, identifying an appropriate charging point, and creating a reservation. The workflow includes checks to ensure that the required vehicle is available and that a usable charging point exists before continuing with the operation. This helps avoid invalid operations and provides a more reliable charging-management process. The reservation information is maintained as part of the application's operational data and can be viewed through the reservation interface.

I contributed to the charging-session management functionality as well. The charging-session module records important information such as the vehicle, charging point, energy consumed, duration, and charging status. I worked on ensuring that the required information is entered and validated before a session is recorded. Once the session is created, the information becomes

available for display in the charging-session module and is also used by other parts of the application, including payment and analytics.

The charging-cost calculation was another important part of my work. The application calculates the charging cost based on the amount of energy consumed and the configured tariff. The formula used is **Charging Cost = Energy Consumed × Tariff per kWh**. The configured tariff in the application is ₹8.50 per kWh. The calculation was tested using the sample charging sessions to verify that the generated cost values were correct. This ensures that the payment information corresponds directly to the energy consumed during a charging session.

I also worked on the payment-record functionality. When a charging session is completed, the corresponding payment information is generated using the calculated charging cost. The payment module provides a structured view of these records and allows the operator to review the amounts associated with completed charging activities. Maintaining this connection between charging sessions and payment information helps the system provide a consistent representation of charging transactions.

Input validation and error handling were also considered during my contribution. The application validates numerical values entered for charging sessions and displays a warning when invalid or non-positive values are provided. The workflow also checks whether a vehicle has been registered before allowing relevant reservation or charging operations to continue. Similarly, the application prevents a charging session from being recorded when there is no usable charging point available. These checks help reduce incorrect data entry and improve the reliability of the application.

I contributed to the integration of operational information with the dashboard and analytics views. The dashboard summarizes important information such as the total number of charging points, available points, occupied points, and energy delivered. The analytics section provides additional information about sessions, energy consumption, revenue, utilization, and charging-point distribution. By connecting these views with the current application records, the system can present operational information in a simple and understandable form. This makes it easier for an operator to monitor the charging environment without manually examining every individual record.

Testing and debugging formed another important part of my contribution. I tested the major workflows of the application, including application launch, dashboard navigation, charging-station management, charging-point status changes, reservation creation, charging-session recording, user and vehicle registration, payment viewing, analytics, and page refresh operations. I also verified the charging-cost calculations and tested invalid input conditions. These activities helped identify workflow issues and confirm that the implemented functionality behaved as expected.

Through this project, I gained practical experience in **Java Swing, object-oriented programming, class and object design, data management, event-driven GUI development, input validation, calculation logic, data visualization, and software testing**. I also developed a better understanding of how different modules of a software application interact with one another. Working on EV ChargeHub helped me apply Java concepts to a practical Smart Campus problem rather than treating each programming concept as an isolated exercise.

Overall, my contribution helped strengthen the functional and operational aspects of the EV ChargeHub system. I was involved in organizing the application's data entities, supporting charging-point and reservation workflows, managing charging sessions and payments, implementing validation, connecting operational information with the dashboard and analytics views, and performing testing and debugging. The supplied prototype demonstrates these functions using structured application objects and local-state handling, providing a foundation that can later be extended with persistent database storage and a dedicated backend. This project improved my technical understanding of Java application development as well as my ability to design, implement, test, and document a complete software solution.